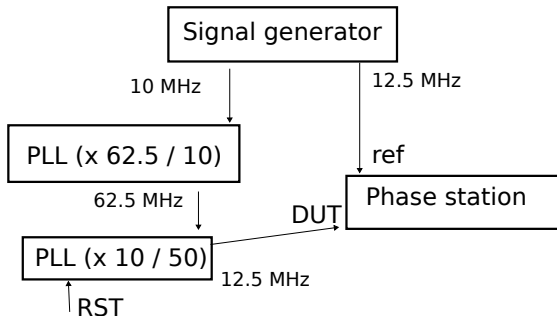
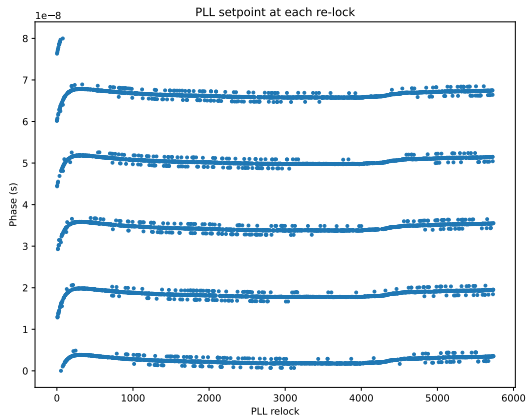


Original Goal: Finding PLLs lock setpoint possibilities

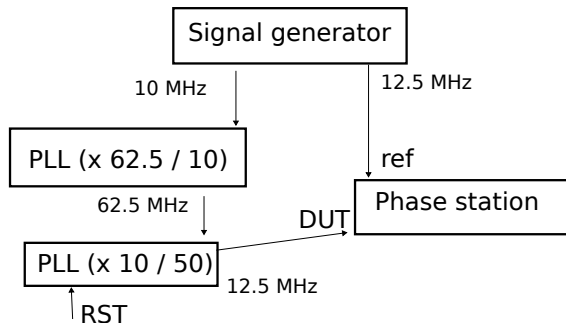
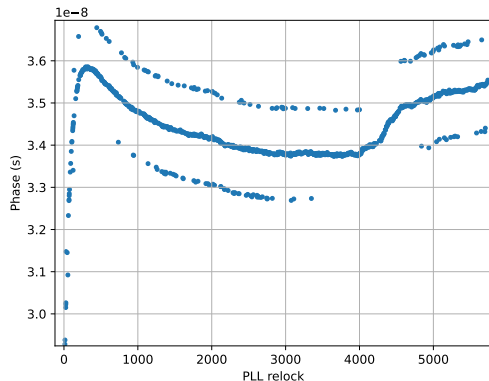
- ▶ PLLs multiply the input by P to VCO frequency and divide the VCO by Q to the output (possibly with a counter)
- ▶ If this counter only takes into account the VCO clock (and not the input), we expect Q possible setpoints.
- ▶ If the PLL is smarter and starts the counter at an input clock edge, it can reduce this to Q/P possible setpoints.



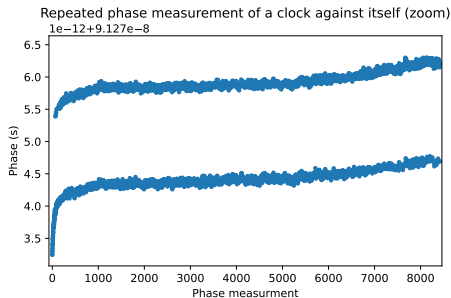
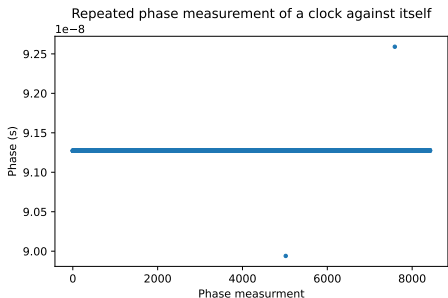
To find what kind of PLLs are implemented in our FPGA we measure the setpoint over multiple resets. The reseted PLL generates 12.5 MHz from 62.5 MHz, with a VCO at 625 MHz. Depending on the PLL implementation, we expect 50 or 5 possible setpoints, respectively 1.6 ns or 16 ns apart.

Finding PLLs lock setpoint possibilities

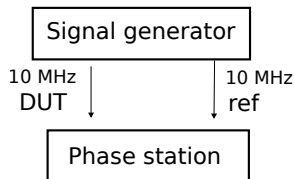
But we find what looks like 15 possible setpoints, 5 main positions 16 ns apart, and 10 others at ± 1.3 ns offsets from the main positions. These ± 1.3 ns offsets are unexplainable by our model of the PLL.



Checking the determinism of the 53100A Phase Noise Analyzer



Since we could not explain these 1.3 ns offsets we checked if they could come from the 53100A Phase Noise Analyzer by feeding it a splitted 10 MHz clock.



We observe (though very rarely) the same 1.3 ns offset on 2 measurements out of over 8k (top-left). We hypothesize that the probability of this issue increases with the clock jitter: the clock from the synthesizer is much cleaner than the one of the FPGA PLLs).

We also notice another (much smaller) indeterministic offset of 1 ps (bottom-left) observed by zooming on the top chart to a 3.5 to 6 ps scale.

Checking the determinism of the 53100A Phase Noise Analyzer

```
const path="../../../Documents/phase_meas";
var acq_num = 0;

function EventRunScript() {
  acq_num = 0;
  run_acq();
}

function run_acq() { // wait for PLL relock
  if (!FileExecuteWait("c:\\...\\mosquitto_sub.exe", " ->
    ↪h 192.168.1.170 -t reset_done -C 1")) {
    ScriptEnd();
  }
  AcqClearParams();
  // Set params with auto generated config
  // [...]
  AcqStartAcquisition("Microchip 53100A ...", true);
}

function EventAcqDone() {
  FileSave(path, "save_" + acq_num + ".tim"); ++acq_num;
  // Send end of measure event to reset the pll
  if (!FileExecute("c:\\...\\mosquitto_pub.exe", " -h →
    ↪192.168.1.170 -t measure_done -m acq_done")) {
    ScriptEnd();
  }
  FileCloseAllPlots();
  run_acq();
}
```

To run these experiments we made this script that is synchronised via MQTT with the computer that hosts the FPGA.

This script waits for a MQTT message signaling that the PLL is locked, then starts the acquisition. When the acquisition is complete it sends a MQTT message, allowing the FPGA to reset the PLL, and goes back to waiting the lock for the next acquisition.

We also tried to make one long acquisition but the phase station doesn't support a clock going off and on again.